

EC-341 Digital System Design

BS (CE)-2021

LAB REPORT #2



Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

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Experiment #2

Introduction to Xilinx Vivado

Objective:

1. Xilinx ISE Design Tool
2. Tasks

Equipment/Software Used:

Xilinx Vivado

Procedure:

Xilinx ISE Design Tool:

Xilinx Tools is a suite of software tools used for the design of digital circuits implemented using Xilinx Field Programmable Gate Array (FPGA) or Complex Programmable Logic Device (CPLD). The design procedure consists of (a) design entry, (b) synthesis and implementation of the design, (c) functional simulation and (d) testing and verification. Digital designs can be entered in various ways using the above CAD tools: using a schematic entry tool, using a hardware description language (HDL) – Verilog or VHDL or a combination of both. In this lab we will only use the design flow that involves the use of Verilog HDL.

The CAD tools enable you to design combinational and sequential circuits starting with Verilog HDL design specifications. The steps of this design procedure are listed below:

1. Create Verilog design input file(s) using template driven editor.
2. Compile and implement the Verilog design file(s).
3. Create the test-vectors and simulate the design (functional simulation) without using a PLD (FPGA or CPLD).
4. Assign input/output pins to implement the design on a target device.
5. Download bit stream to an FPGA or CPLD device.
6. Test design on FPGA/CPLD device

A Verilog input file in the Xilinx software environment consists of the following segments:

Header: module name, list of input and output ports.

Declarations: input and output ports, registers and wires.

Logic Descriptions: equations, state machines and logic functions.

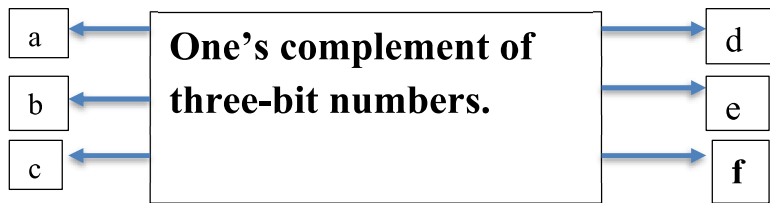
End: end module

Tasks

Task #1:

Write a Gate Level Verilog code to find 1's complement of three-bit numbers.

Block Diagram:



Block diagram of One's complement of three-bit numbers.

Truth Table:

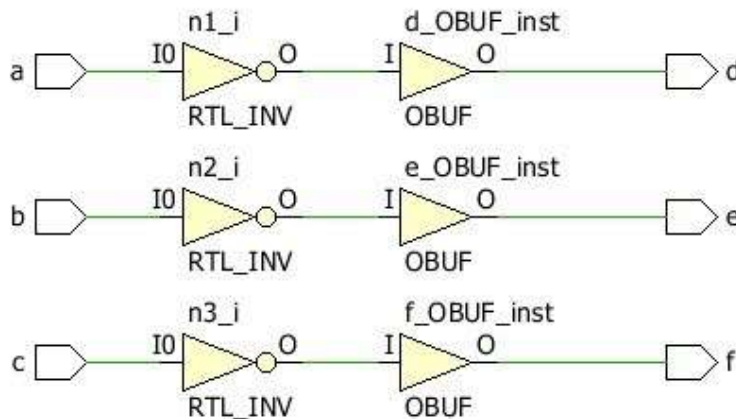
Truth table of one's complement of three-bit numbers

Input			Output		
a	b	c	d	e	f
0	0	0	1	1	1
0	0	1	1	1	0
0	1	0	1	0	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	0	1	0
1	1	0	0	0	1
1	1	1	0	0	0

Boolean Expression:

Simplify:

Logisim:



Verilog:

```

21 module complement(
22     input a,
23     input b,
24     input c,
25     output d,
26     output e,
27     output f
28 );
29     not n1(d,a);
30     not n2(e,b);
31     not n3(f,c);
32 endmodule

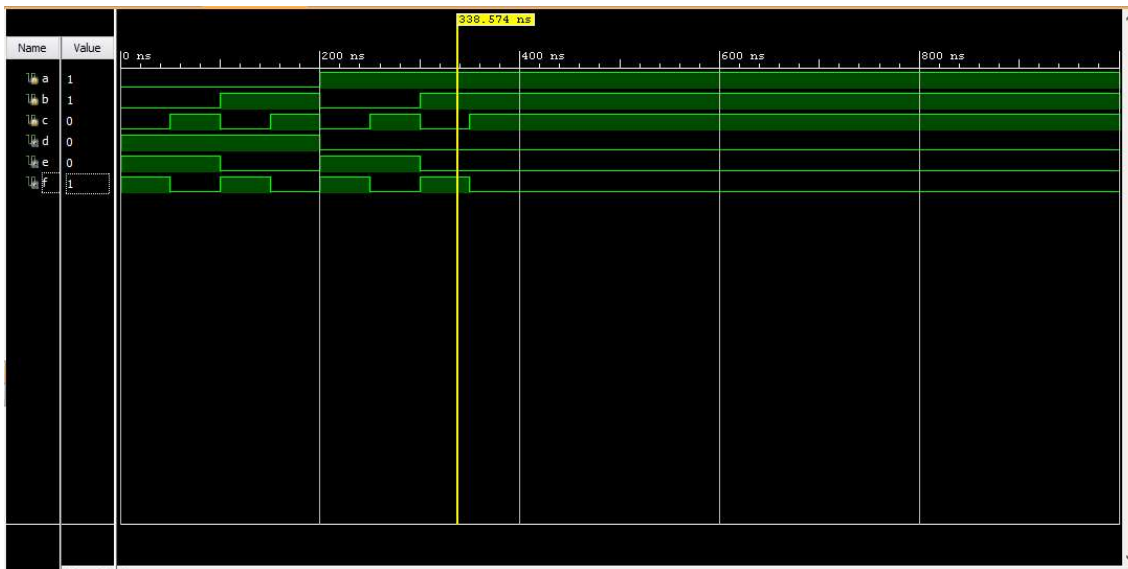
```

```

21 module comptb();
22     reg a,b,c;
23     wire d,e,f;
24     complement obj(a,b,c,d,e,f);
25     initial
26     begin
27         a=0;b=0;c=0;
28         #50 a=0;b=0;c=1;
29         #50 a=0;b=1;c=0;
30         #50 a=0;b=1;c=1;
31         #50 a=1;b=0;c=0;
32         #50 a=1;b=0;c=1;
33         #50 a=1;b=1;c=0;
34         #50 a=1;b=1;c=1;
35     end
36 endmodule

```

Waveform:



Waveform of one's complement of three-bit numbers

Task #2:

Write a Gate Level Verilog code for design and test bench of Half Adder in Xilinx Vivado.

Simulate the Adder for different set of inputs. Attach waveform screenshots, schematic and Verilog code.

Block diagram:



Block diagram of halfadder

Truth Table:

Truth table of halfadder

A	B	Sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Boolean Expression:

Sum: $A \oplus B$

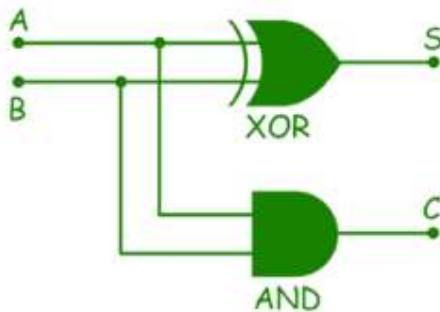
Carry: $A \& B$

Simplify:

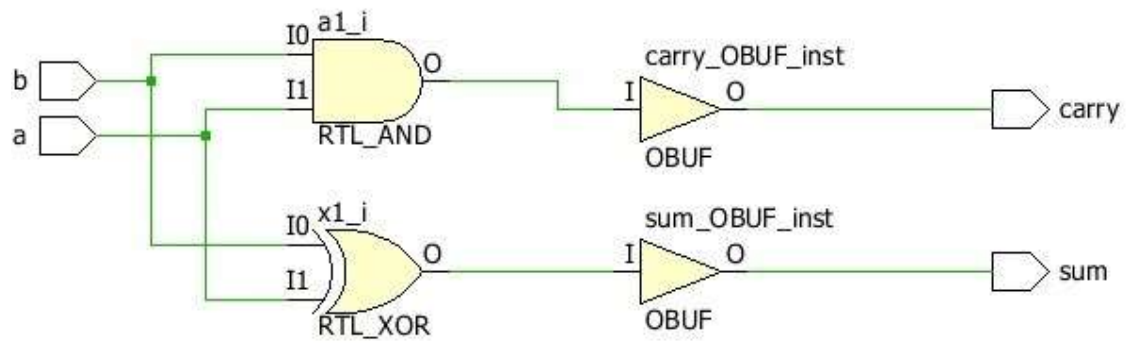
Sum: $A \oplus B (A+B)$

Carry: $A \& B (A.B)$

Logisim:



Schematic diagram:



Verilog:

```

23 module halfadder(
24     input a,
25     input b,
26     output sum,
27     output carry
28 );
29 xor x1(sum,a,b);
30 and a1(carry,a,b);
31 endmodule

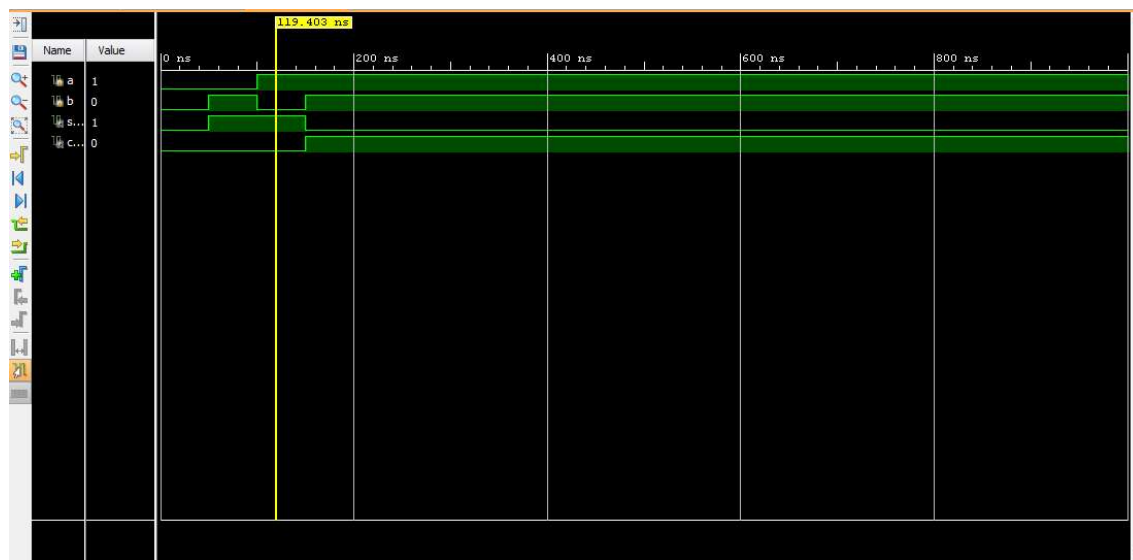
```

```

21 module testhalfadder();
22 reg a;
23 reg b;
24 wire sum;
25 wire carry;
26 halfadder obj(a,b,sum,carry);
27 initial
28 begin
29 a=0;b=0;
30 #50 a=0;b=1;
31 #50 a=1;b=0;
32 #50 a=1;b=1;
33 end
34 endmodule

```

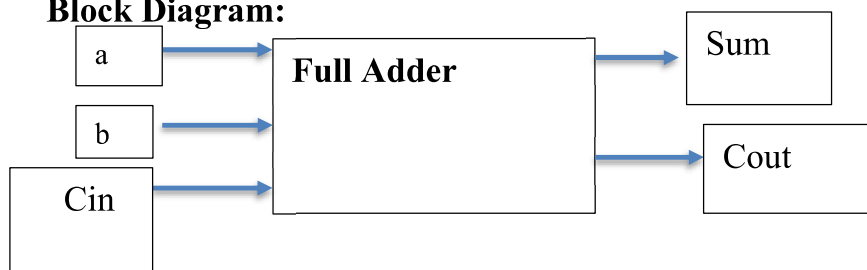
Waveform:



Task #3:

Write code for above task for Full Adder as well.

Block Diagram:



Block diagram of full adder

Truth Table:

Truth table of full adder

Input			Output	
A	B	C _{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Boolean Expression:

$$\text{Sum} = A \oplus B \oplus C_{in}$$

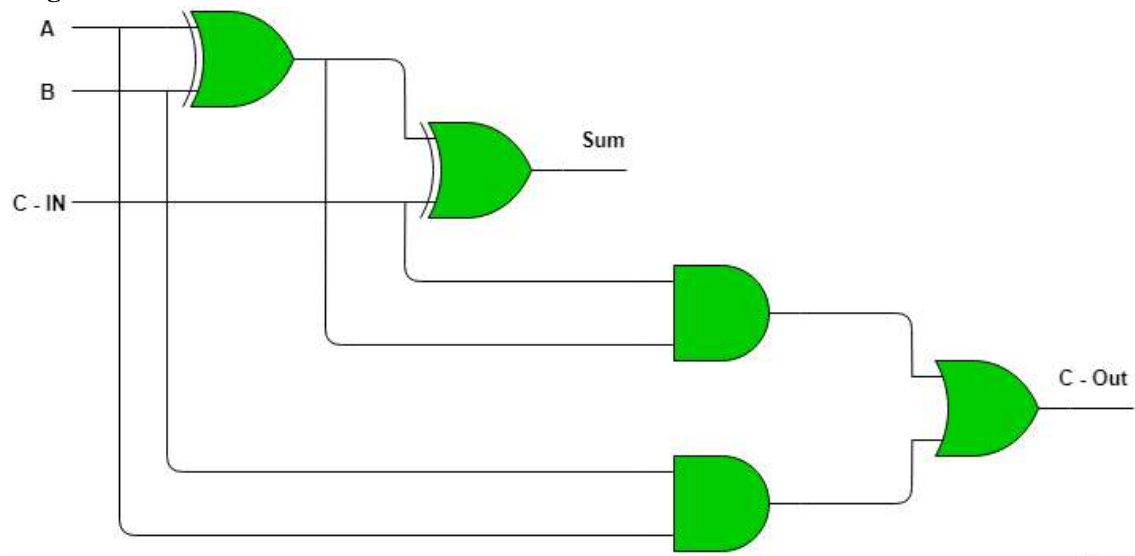
$$\text{Cout} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$$

Simplify:

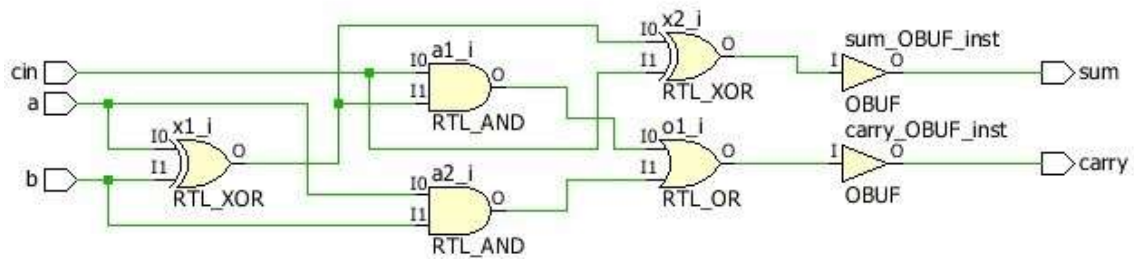
$$S = A \oplus B \oplus C_{in}$$

$$\text{Cout} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$$

Logisim:



Schematic diagram:



Verilog:

```
21 module fa(
22     input a,
23     input b,
24     input cin,
25     output sum,
26     output carry
27 );
28     wire w1,w2,w3;
29     xor x1(w,a,b);
30     xor x2(sum,cin,cin);
31     and a1(w2,cin,w1);
32     and a2(w3,a,b);
33     or o1(carry,w2,w3);
34 endmodule

23 module fatestbench();
24 reg a,b,cin;
25 wire carry,sum;
26 fa obj(a,b,cin,sum,carry);
27 initial
28 begin
29 a=0;b=0;cin=0;
30 #50 a=0;b=0;cin=1;
31 #50 a=0;b=1;cin=0;
32 #50 a=0;b=1;cin=1;
33 #50 a=1;b=0;cin=0;
34 #50 a=1;b=0;cin=1;
35 #50 a=1;b=1;cin=0;
36 #50 a=1;b=1;cin=1;
37 end
38 endmodule
```

Waveform:**Conclusion:**

In this lab, we have learnt the Xilinx vivado and Verilog. Also, we design the halfadder, full adder and one's complement of three-bit numbers on gate level.